

Types of Databases

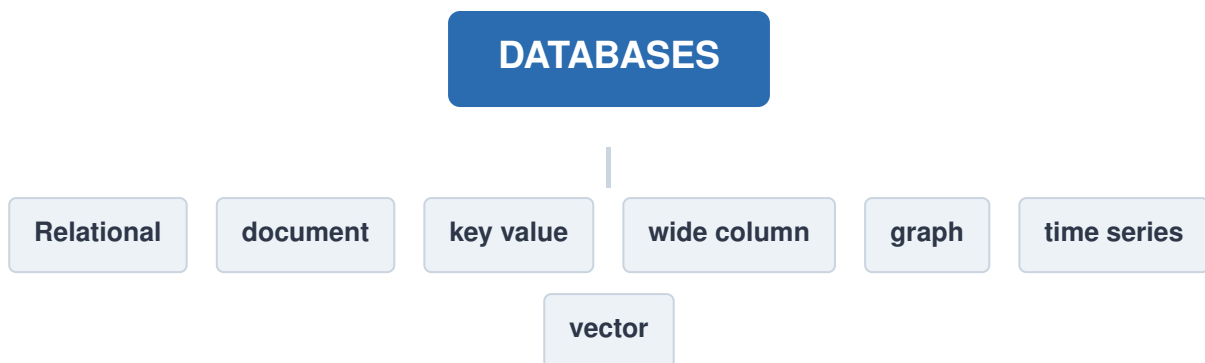
Mongo DB, redis, cassandra wont be discussed here, It comes later in Phase 26 (NOSQL)

different databases for applications have different requirements :-

- Structured data
- fast caching
- complex relation
- time based measure
- AI embeddings



hence different database models evolved



1. Relational database

My SQL, postgre SQL, oracle database

data is stored in tables, rows are called tuples columns are called attributes , relations are maintained using foreign keys

FEATURES

- Structured schema
- SQL support
- ACID transactions
- Joins
- Normalization
- Constraints

CHARACTERISTICS

first choice when data integrity is critical

used in Banking, Ecomm, ERP, inventory, hospital systems etc

Cons:

less flexible schema, horizontal scaling challenging

2. Document Database

Mongo DB, couch DB

stores data as documents, JSON or BSON typically. each document can have a different structure

```
{
  "studentId" : 006,
  "name" : "Aady",
  "Skills" : [ "VLSI", "RISC-V", "C" ]
}
```

CHARACTERISTIC :

- flexible schema
- nested objects
- JSON based
- easy to evolve

BEST FOR

user profiles, blogging platforms, rapidly changing applications

Cons:

data duplication is common, complex joins are weaker than relational

3. Key - Value database

Redis, Amazon dynamo DB

a giant hash map, where we ask for key and we get its value.

fast lookups, memory based, simple model

BEST FOR :

caching, sessions, rate-limiting, shopping cart, leader board

PROS → simple ops, low latency, high throughput

CONS → no joins, limited querying

4. Wide column database

Apache Cassandra, Apache HBase

these are optimized for large scale distributed workload
unlike relational, rows can have different columns

massive scalability, distributed by design, handles huge datasets

Roll	Name	email	phone
006	Aadya	NULL	79870...
012	Arpit	hehe@gmail.com	NULL

BEST FOR

logging, social media activity feed, telecom, IoT

pros → excellent horizontal scaling, high throughput, fault tolerant

cons → complex, few relational features

5. Graph database

Neo4j, Amazon neptune

Stores nodes and relationships (edges) instead of tables, relationships are stored directly



BEST FOR

knowledge graphs, recommendation systems, fraud detection, network analysis

PROS

efficient relationship traversal, natural representation of connected data

CONS

not ideal for traditional transactional business application

6. Time - Series Database

influx DB, timescale DB

stores data ordered by time, Eg Timestamp vs temperature

BEST FOR :

sensors, IoT, server monitor, metrics collection

PROS

efficient time based queries, data compression

CONS

specialized for temporal workloads rather than general purpose

7. Vector database

Pinecone, Weaviate

instead of storing rows, it stores embedding and enable similarity search

BEST FOR

AI chatbots, RAG, recommendation system, semantic search

PROS

semantic retrieval, fast nearest neighbour search

CONS

not a replacement, used alongside SQL

Practice Questions

Q1. Why do different types of databases exist instead of one universal database?

A1: Different database types exist because workloads differ significantly. Transaction-heavy systems require strong consistency and relational modeling, while caching, graph traversal, time-based analytics, or AI similarity search each benefit from specialized storage models and query mechanisms.

Q2. Compare relational and document databases. Give one use case where each is the better choice.

A2: A relational database stores structured data in tables with predefined schemas and is ideal for applications such as banking or order processing where ACID transactions and joins are essential. A document database stores flexible JSON-like documents, making it better suited for evolving user profiles, content management systems, or product catalogs.

Q3. A company needs to store millions of user login sessions with very fast retrieval by session ID. Which database type would you recommend and why?

A3: A key-value database is the best choice. Login sessions are naturally modeled as session_id → session_data mappings, requiring extremely fast reads and writes without complex relational queries.

Q4. Why are graph databases well suited for social networking applications?

A4: Graph databases store entities as nodes and relationships as edges, enabling efficient traversal of connected data such as friends, followers, or recommendations without relying on multiple expensive joins.

Q5. An IoT company records temperature readings from one million sensors every second. Which database type is most appropriate? Explain.

A5: A time-series database is most appropriate because the data is timestamped, append-heavy, and commonly queried over time ranges for aggregation, monitoring, and trend analysis.

Q6. What is a vector database, and why has it become important in modern AI applications?

A6: A vector database stores high-dimensional embeddings produced by machine learning models and supports efficient similarity search. It is a core component of modern AI systems such as semantic search, recommendation engines, and Retrieval-Augmented Generation (RAG), where finding the most semantically similar data is more important than exact keyword matching.